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Institut für Klinische Genetik
Universitätsklinikum Carl Gustav Carus Dresden
Ref. 1943

Re: Bioinformatiker*in — Genomsequenzierung in der Onkologie (Ref. 1943)

Dear Hiring Committee,

I am applying for the bioinformatician position (Ref. 1943) in the Institute for Clinical Genetics, supporting cancer genome sequencing diagnostics and therapy identification. The position description maps precisely to the software and pipelines I have built independently: genome-wide variant analysis at production scale, RNA-seq integration, machine learning applied to multidimensional cancer genomic data, and large-scale omics pipelines with API-based database integration.

NGS data analysis and cancer genome sequencing pipelines. The Gospel framework is a production-grade genomic analysis system processing genome-wide variant data at 10^6 variants per second. The pipeline covers the full NGS analysis workflow: VCF processing and variant filtering, structural variant detection, population frequency annotation against gnomAD v3.1.2, 1000 Genomes Phase 3, and UK Biobank, RNA-seq quantification and differential expression, functional annotation via pathway databases (KEGG, Reactome), and protein structure analysis (RDKit). For the oncology context specifically, Gospel implements somatic variant analysis: distinguishing tumour-specific alterations from germline variants using population frequency thresholds and allele frequency in tumour-normal pairs, filtering against cancer hotspot databases, and annotating variants with predicted functional consequences. The pipeline is API-integrated against the major reference databases — gnomAD, ClinVar, COSMIC, OMIM — using structured query patterns designed for the kind of automated NGS data flows the position describes. All processing is reproducible, version-controlled, and runs on distributed computing infrastructure (Ray, Docker) scalable to clinical laboratory volumes.

Machine learning for multidimensional cancer genomic data. The position lists ML applied to multidimensional biological data as a core requirement. I have built three directly relevant systems:

The Syndrome neoantigen predictor identifies which tumour somatic mutations produce immunogenic neoantigens from the combination of MHC presentation likelihood ($\rho_{\mathrm{MHC}} \geq 0.72$) and self-distinction ($\Delta\rho_{\mathrm{self}} > 0.15$), using spectral DFT on three physicochemical channels with $\mathcal{O}(cL \log L)$ complexity. For cancer genomics, this converts a tumour mutational burden VCF directly into a ranked neoantigen list without requiring epitope database lookup — a pipeline step relevant to immunotherapy candidate selection in the institute's oncology programme.

The resistance predictor applies spectral complementarity ρ to predict drug-enzyme resistance from the enzyme sequence and drug structure: susceptibility when $\rho \geq \rho^* = 0.50$. Validated against clinical resistance positions in TEM-1/Ampicillin, GyrA/Ciprofloxacin, and AAC(3)-IV/Gentamicin. For personalised cancer therapy identification, this provides a first-pass computational screen of patient-specific enzyme variants for drug resistance before clinical testing.

The disease state model represents patient state as $\mathbf{D} \in [0,1]^8$ across eight pathophysiological dimensions with per-dimension coherence indices; $\mathcal{O}(N \log N)$ trajectory computation; sparse ℓ_1 LP for minimum-intervention therapy selection. AUC $= 1.00$ in 25/25 loop-holonomy self-consistency validation experiments. This provides the therapy-finding layer the position describes: given the patient's genomic and clinical data, identify the minimum intervention set to redirect disease trajectory to the healthy attractor.

Pipeline maintenance, validation, and database integration. The position emphasises maintaining existing bioinformatic data flows and integrating complex databases via API queries. My experience here is direct: Gospel integrates against gnomAD, ClinVar, OMIM, COSMIC, KEGG, and Reactome through structured REST and GraphQL API clients; Lavoisier integrates against LIPID MAPS, LipidBlast, HMDB, and the NIST mass spectral library for automated compound annotation. In my TUM lipidomics work I maintained and extended multi-site federated MS analysis pipelines across distributed clinical computing environments, including version control of pipeline configurations, automated validation of output against reference standards, and documentation for non-bioinformatics users. For software validation specifically, all frameworks I have built are tested against reference databases with documented accuracy metrics — 98.9 % NIST feature extraction for Lavoisier, AUC $= 0.81$ vs NetMHCpan for MHC binding prediction, 10^6 variants/second for Gospel — providing the kind of quantified validation the clinical genetics context requires.

Background. I hold an MSc in Computational Biology (Universität Tübingen) covering sequence analysis, variant calling, population genetics, and bioinformatics pipeline development; and I conducted doctoral research in Computational Lipidomics at TUM working on large-scale MS-based molecular profiling and multi-omics integration. I also have direct cancer biology experience: UPLC-MS/MS shotgun proteomics for cancer biomarker discovery at the University Medical Center Groningen (2013). I work primarily in Python, with Rust for performance-critical components; Java and Bash from prior work. I am legally resident in Germany with full work authorisation and available immediately. English is native; German is conversational. I am willing to relocate to Dresden.

Yours sincerely,

Kundai Farai Sachikonye